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01 ADMIN DASHBOARD SPECIFICATION

The admin-service is a separate FastAPI application accessible only from within the VPC (no public internet exposure). Admin users authenticate via SSO (Google Workspace). Role-based access control (RBAC) with 4 roles: Super Admin, Trust & Safety, Verification Agent, Analyst. All admin actions are audit-logged to an immutable PostgreSQL table with actor, action, timestamp, IP.

Admin Roles & Permissions

Role	Moderation	Verification	User Mgmt	Analytics	System Config	AI Prompts
Super Admin	✓ Full	✓ Full	✓ Full	✓ Full	✓ Full	✓ Full
Trust & Safety	✓ Full	✓ View	✓ Suspend/Ban	✓ Full	✗	✗
Verification Agent	✓ View	✓ Full	✓ View	✓ View	✗	✗
Analyst	✓ View	✓ View	✓ View	✓ Full	✗	✗

Module 1: Moderation Queue

All newly created profiles enter this queue before going live (Tier-1 review). Target SLA: 48 hours. Escalated flags: 4-hour SLA.

Profile Review Screen	
Queue View	Paginated list: profile photo thumbnail, name, city, religion, caste, submitted_at, queue_age (hrs), flag count
Sort/Filter	By: oldest first (default), flag type, religion, state, verification tier, age bracket
Bulk Actions	Approve all / Reject all / Mark for human review — with confirmation modal
Review Pane	Side-by-side: profile data (all fields) AI flags from Rekognition + Claude decision buttons
Decision Options	Approve → Tier-1 Reject (with reason) Request re-upload Escalate → Senior
Rejection Reasons	Explicit photo Multiple faces Celebrity/fake Incomplete data Underage Profanity Spam
Audit Trail	Every decision auto-logged: agent_id, action, reason, timestamp, profile_id
Appeals	Rejected users can appeal once; appeal enters separate queue with senior agent assignment

Photo Moderation Sub-Screen	
Rekognition Flags	Explicit: Sexual/Suggestive Violence Visually Disturbing Rude Gestures — shown with confidence %
Face Count Alert	Photos with >1 face flagged in red. Agent must confirm primary face or reject.
Quality Score	0-100 AI quality rating (lighting, blur, resolution). Below 40 → recommend re-upload.
Actions	Approve photo Reject photo (keep profile) Reject profile Set as primary Re-order
Video Profile	First-frame Rekognition scan + manual 30-sec video review by agent for content safety

Module 2: Verification Review Queue

Tier-2 (Blue) verification requests land here after automated Textract + Rekognition checks. Human agent makes final approval decision within 24hr SLA.

Tier-2 Verification Review Screen	
Queue View	Profile ID Submitted_at Document type Textract match status Rekognition face match % Agent assigned
Document Viewer	Side-by-side: uploaded ID (S3 pre-signed URL, expires 1hr) Extracted fields from Textract
Textract Results	Name extracted DOB extracted Match vs profile: ✓ Match / ■ Partial / ✗ Mismatch (highlighted)

Selfie Comparison	LiveKit liveness result Rekognition CompareFaces similarity % Both images displayed
Decision Threshold	Auto-approve if: Textract name match ≥ 90% AND Rekognition similarity ≥ 80% AND liveness PASS
Manual Override	Agent can approve/reject overriding AI thresholds with mandatory justification note
Rejection Reasons	Name mismatch DOB mismatch Expired document Unclear photo Liveness fail Suspected fake
Tier-3 Upgrade	Gold verification: agent reviews employment letter / address doc / background check API result
SLA Breach Alert	Queue items > 20 hours → PagerDuty alert to Trust & Safety lead

Module 3: User Management

User Search & Profile Management	
Search	By: phone, email, matrimony ID, name, IP address, device fingerprint
Profile View	All profile fields Photos Videos Verification status Subscription Interaction history summary
Activity Log	Login history (IP, device, time) Message count Report count Block count Flag count
Actions — Soft	Send warning email Force profile re-review Remove specific photo Downgrade verification
Actions — Hard	Suspend (1/7/30/permanent) Ban device fingerprint Ban phone number Ban email domain
Scam Investigation	View flagged messages (decrypted view requires 2-agent approval + audit log entry)
Bulk Operations	Export users by filter (CSV) Bulk notify Bulk suspend (requires Super Admin)
Unban Requests	Users can request unban after 30 days; separate queue with Super Admin decision

Module 4: Reports & Flags Dashboard

Report Investigation Screen	
Report Queue	All user reports sorted by: newest, most-reported profile (past 7 days), severity
Report Categories	Fake profile Harassment Explicit content Scam/fraud Spam Underage Impersonation
Reporter Context	Shows reporter's profile, their interaction history with reported user, screenshot if provided
Reported User Context	Their profile, prior reports received, current suspension status, match history
Decision Options	No action (dismiss) Warning issued Photo removed Profile suspended Profile banned
Auto-Escalation	5 reports in 7 days from different users → auto-escalate to Trust & Safety immediately
Scam Alert Review	Messages flagged by Claude scam detector appear here with risk level + flagged phrases highlighted
Pattern Detection	Dashboard shows: top reported phrases, top reported cities, spike alerts (>3 reports/day/profile)

Module 5: Analytics Dashboard

Grafana-based dashboards embedded in admin panel. Read-only for Analyst role. Data refreshed every 15 minutes via CloudWatch metrics pipeline.

Dashboard 1: Platform Health (Real-Time)	
Active Users	Current WebSocket connections DAU trend (7-day sparkline) MAU counter
API Health	Request rate P50/P95/P99 latency Error rate by service Slow query count
Queue Depth	Moderation queue count Verification queue count Notification queue lag
Claude API	Calls/minute Token usage/hour Error rate Cost today vs budget

Infrastructure	ECS task health RDS CPU/IOPS Redis memory Kafka consumer lag
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Dashboard 2: Business Metrics (Daily/Weekly)

Registrations	Daily new signups By source (organic/referral/ad) By state/city Funnel: signup→profile complete
Engagement	DAU/MAU ratio Average session duration Profiles viewed per session Like/skip ratio
Matching	Match score distribution Average matches/user Mutual match rate Match-to-message rate
Messaging	Messages sent/day Average conversation length Response rate VoIP minutes/day
Conversion	Free→Premium rate Premium→Platinum rate Churn rate MRR ARPU LTV estimate
Telugu Specifics	Telugu users % AP/TS vs NRI ratio Top communities by activity City-wise heatmap
AI Performance	Bio generation uses/day Match explanation views Scam blocks/day Icebreaker acceptance rate

Dashboard 3: Trust & Safety Metrics

Verification	Tier-1 approval rate Tier-2 approval rate Average review time Rejection reasons distribution
Reports	Reports filed/day Resolution time Action taken distribution Repeat offender rate
Scam Detection	Claude flags/day Block rate False positive rate (from user appeals) Top scam phrases
Fake Profiles	Detected/day Detection method (AI vs user report) Time-to-detection average
SLA Compliance	Moderation queue: % within 48hr Verification queue: % within 24hr Escalation: % within 4hr

Module 6: Content & AI Management

Prompt Management	View/edit Claude prompt templates by feature. Version history. A/B test two prompts (50/50 split).
Prompt Testing	Send test inputs to Claude Sonnet/Opus directly from admin panel. View raw JSON response.
Scam Phrase Library	Editable list of known scam phrases fed into Claude system prompt. Add/remove phrases.
Caste Data Management	Add/edit/merge caste entries, sub-caste entries, gothra entries. Flag duplicates.
City/Region Data	Add new cities, map to districts/states. Telugu city canonical names + aliases.
Notification Templates	Edit push/email/SMS copy templates. Preview on mock device. Deploy with rollback.
Feature Flags	Enable/disable features per user segment without code deploy (LaunchDarkly or custom).
ML Model Monitor	Current XGBoost model version Training date Accuracy metrics Manual retrain trigger

02 PUSH NOTIFICATION TAXONOMY

All notifications delivered via AWS SNS → Firebase Cloud Messaging (Android) and APNs (iOS). Email via Amazon SES. SMS via AWS SNS (only for critical auth events). Users can manage opt-in/opt-out per category in Settings → Notifications. Global daily cap: 8 push notifications per user per day. Quiet hours: 22:00–08:00 local time (no push).

Notification Categories & Opt-Out

Category	Channel	Default	User Can Opt-Out	Daily Cap
Auth & Security	SMS + Push + Email	ON	Partial (security cannot be off)	No cap
Matches & Recommendations	Push + Email	ON	Yes (full)	2/day
Interactions (likes/views/interests)	Push	ON	Yes (full)	3/day
Messages	Push	ON	Yes (mute per chat or all)	No cap (per message)
Saved Search Alerts	Email	OFF (user enables)	Yes	1/day per search
Profile Completion Nudges	Push + Email	ON	Yes	1/day
Subscription & Billing	Email	ON	Cannot opt-out (legal)	As needed
Success Stories & Community	Email + Push	OFF	Yes	2/week
Promotional & Offers	Push + Email	OFF	Yes	3/week
System & Maintenance	Email	ON	Cannot opt-out	As needed
Match Hour Events	Push	OFF	Yes	1/event

Complete Notification Type Reference

Auth & Security Notifications			
Event ID	Channel	Copy	Trigger
OTP_SEND	SMS	Your OTP is {otp}. Valid for 5 minutes. Do not share with anyone.	Trigger: OTP request
LOGIN_NEW_DEVICE	Push+Email	New login detected from {device} in {city}. Not you? Secure your account now.	New device login
PASSWORD_CHANGED	Email	Your password was changed on {datetime}. Contact support if this wasn't you.	Password update
ACCOUNT_SUSPENDED	Email+Push	Your account has been suspended. Reason: {reason}. Appeal within 7 days.	Admin suspension
ACCOUNT_BANNED	Email	Your account has been permanently deactivated. {reason}	Admin ban
VERIFICATION_APPROVED	Push+Email	Congratulations! Your {tier} verification is approved. Your profile now has the Verified badge.	Tier 2/3 approval
VERIFICATION_REJECTED	Push+Email	Your {tier} verification could not be approved. Reason: {reason}. Try again anytime.	Tier rejection
SCAM_WARNING	Push	Safety alert: A message you received contains a potential scam pattern. We've flagged it for your protection.	Claude scam flag
Matches & Recommendations			
Event ID	Channel	Copy	Trigger
DAILY_MATCHES_READY	Push	Your {count} new matches for today are ready! {top_name} from {city} looks like a great fit.	Daily at 9 AM local
MUTUAL_MATCH	Push	It's a Mutual Match! You and {name} both like each other. Start a conversation now!	Bidirectional like
JUST_JOINED_ALERT	Push	{count} new profiles matching your search joined today. Be among the first to connect!	New profiles in filters

Event ID	Channel	Copy	Trigger
SAVED_SEARCH_ALERT	Email	Based on your saved search "{search_name}": {count} new profiles match today.	Daily/weekly digest
HIGH_MATCH_ALERT	Push	We found a 90%+ compatibility match for you: {name}, {age}, {city}. Don't miss this one!	Score ≥ 90
WEEKLY_DIGEST	Email	Your week on Matrimony: {views} profile views, {likes} likes received, {matches} new matches.	Sunday 10 AM
PROFILE_VIEWED_MILESTONE	Push	{count} people viewed your profile this week! Upgrade to see who they are.	Milestone: 10/50/100

Interactions — Likes, Views, Interests

Event ID	Channel	Copy	Trigger
LIKE_RECEIVED	Push	{name} liked your profile! View their profile and connect.	Profile liked
SUPER_LIKE_RECEIVED	Push	■ {name} sent you a Super Like! They're really interested in you.	Super like received
INTEREST_RECEIVED	Push	{name} sent you an interest request{msg_preview}. Accept to start chatting.	Interest sent
INTEREST_ACCEPTED	Push	Great news! {name} accepted your interest. You can now chat with them!	Interest accepted
INTEREST_DECLINED	Push	{name} has declined your interest at this time.	Interest declined (gentle)
SHORTLISTED	Push	{name} shortlisted your profile. Make your move!	Profile shortlisted
PROFILE_VIEW_TEASER	Push	{count} people viewed your profile today. Upgrade to Premium to see who!	Free user: daily count
VIRTUAL_GIFT_RECEIVED	Push	{name} sent you a {gift_type}! Open to see their message.	Gift received

Messages & Calls

Event ID	Channel	Copy	Trigger
NEW_MESSAGE	Push	{name}: {message_preview_20chars}	New chat message
MESSAGE_UNREAD_REMINDER	Push	You have {count} unread messages. {name} is waiting for your reply.	Unread after 4hrs (1x only)
FIRST_MESSAGE	Push	{name} sent you their first message! They said: "{preview}". Don't leave them waiting!	First ever message
ICEBREAKER_ACCEPTED	Push	Your icebreaker worked! {name} replied to your message. Keep the conversation going!	Reply after icebreaker
CALL_MISSED	Push	You missed a call from {name}. Tap to call back.	Missed VoIP call
CALL_VOICEMAIL	Push	{name} left you a voice message.	Voice message left
GIFT_IN_CHAT	Push	{name} sent you a {gift_type} in chat! ■	Gift in conversation

Profile Completion & Billing

Event ID	Channel	Copy	Trigger
PROFILE_50_PCT	Push	You're halfway there! Complete your profile to appear in search results and get more matches.	Completion hits 50%
PROFILE_75_PCT	Push	75% done! You're now visible in search. Add more photos to boost your match rate by 2x.	Completion hits 75%
PROFILE_INCOMPLETE_NUDGE	Push	Your profile has been inactive 3 days. Complete it to avoid missing matches!	3 days at <75%, 1x only
PHOTO_REJECTED	Push	One of your photos was removed during review. Upload a clear, solo face photo to improve your profile.	Photo moderation reject
VIDEO_PROFILE_PROMPT	Push	Add a 30-second intro video to your profile — users with videos get 2x more views!	No video after 7 days
SUBSCRIPTION_EXPIRING	Push+Email	Your Premium subscription expires in {days} days. Renew now to keep all your benefits.	3 days before expiry
SUBSCRIPTION_EXPIRED	Push+Email	Your Premium subscription has expired. Upgrade again to restore all features.	Day of expiry

Event ID	Channel	Copy	Trigger
SUBSCRIPTION_RENEWED	Email	Your Premium subscription has been renewed for {period}. Amount: ■{amount}.	Auto-renewal success
PAYMENT_FAILED	Push+Email	Your subscription renewal failed. Update your payment method to avoid losing access.	Payment failure
BOOST_LIVE	Push	Your Profile Boost is now active! You're featured at the top of search results for the next 24 hours.	Boost activated
SPOTLIGHT_LIVE	Push	Your Spotlight is live for the next hour! You're featured on the homepage. ■	Spotlight activated
CREDITS_LOW	Push	You have only {count} gift credits left. Top up to keep sending gifts!	Credits < 5
FREE_PREMIUM_REWARD	Push+Email	You've earned 1 month of Premium free! A friend you referred just subscribed. Thank you!	Referral reward

Community & Engagement

Event ID	Channel	Copy	Trigger
MATCH_HOUR_REMINDER	Push	Telugu Match Hour starts in 30 minutes! Join now for 2× profile visibility.	30 min before event
MATCH_HOUR_LIVE	Push	Telugu Match Hour is LIVE! Connect with profiles who are online right now.	Event start
SUCCESS_STORY_FEATURED	Push	A couple from your community just shared their success story. Read how they found each other! ♥■	New story published
FESTIVAL_BOOST	Push	Ugadi Special: Boost your profile for free today only! A new beginning awaits.	Telugu festival day
ANNIVERSARY_REMINDER	Push	Today is special — it's been {months} months since you joined. Update your profile to keep it fresh!	Monthly milestone
RESPONSE_RATE_BADGE	Push	You've earned the 'High Responder' badge! Profiles with this badge get 30% more interest requests.	Response rate >70%
REFERRAL_JOINED	Push	Your friend {name} just joined using your referral! They need to subscribe for you to earn your free month.	Referral registers

Frequency Cap Implementation

Redis sorted sets per user per category. Score = timestamp. Count items in last 24hr window. If count ≥ cap → enqueue for next day (not silently dropped). Quiet hours enforced via user timezone stored in profile.

Global daily cap	8 push notifications per user per 24-hour rolling window (Redis counter)
Quiet hours	22:00–08:00 user's local time. Notifications queued, delivered at 08:01.
Per-category cap	Enforced in addition to global cap. Whichever limit hit first wins.
Unread message rule	Message push: max 1 per conversation per 5 minutes (prevent spam on active chat)
Reminder dedup	Completion nudge, subscription expiring: send once only per trigger event (Redis dedup key, 7-day TTL)
Opt-out immediate	Opt-out respected immediately (Redis flag checked before send). No 'one more' sends.
Unsubscribe tracking	All email unsubscribes processed via SES Suppression List within 10 minutes.

03 THIRD-PARTY VENDOR LIST

All vendors must sign a Data Processing Agreement (DPA) before integration. DPDP-compliant DPA templates to be prepared by legal counsel. Vendors storing Indian personal data must comply with DPDP Act 2023. Vendor risk tier: Critical (P0) = platform cannot function without it; Important (P1) = major feature loss; Supplementary (P2) = degraded experience only.

Critical (P0) — Platform Cannot Function Without

Vendor	Purpose	Pricing Model	Est. Monthly Cost	Where to Sign Up	Critical Action
AWS (Amazon Web Services)	Cloud infrastructure (compute, DB, storage, CDN, messaging)	Pay-as-you-go + Reserved	~\$800-35K/month	accounts.aws.com — Sign up + enable Business Support (\$100/month recommended)	Contact AWS SA for startup credits (up to \$25K)
Anthropic	Claude Sonnet 4.6 + Opus 4.6 API	Pay-per-token	~\$300-12K/month	console.anthropic.com — API key creation	Apply for higher rate limits before launch (default 60 RPM)
Razorpay	Payment gateway — subscriptions + one-time	1.5-2% per transaction + 0 setup	% of revenue	dashboard.razorpay.com — KYC + bank account	Sign Razorpay T&C; + complete KYC within 3 business days
Firebase (Google)	FCM push notifications (Android)	Free tier: unlimited	\$0 (under quota)	console.firebase.google.com — Create project + add SHA-1	Add google-services.json to React Native build
Apple Developer Program	APNs push (iOS) + App Store distribution	\$99/year	\$99/year	developer.apple.com — Enroll as organization	Requires D-U-N-S number (allow 5 business days)
Google Play Console	Android app distribution	\$25 one-time	\$25	play.google.com/console — Developer account	One-time registration; merchant account separate for IAP

Important (P1) — Major Feature Loss

Vendor	Purpose	Pricing Model	Est. Monthly Cost	Where to Sign Up	Critical Action
LiveKit	WebRTC VoIP audio+video calls, number masking	Self-hosted on EC2 OR LiveKit Cloud	\$200-2K/month (cloud) or \$0 (self-hosted)	livekit.io — Cloud account OR deploy open-source server	Recommend self-hosted EC2 for cost control at scale
Pinecone	Vector DB for semantic profile matching	Starter: free. Standard: \$70+/month	\$70-500/month	app.pinecone.io — Create account + index	Create index: dimension=1536, metric=cosine
AWS Rekognition	Photo moderation, face detect, liveness	\$0.001-0.01/API call	\$50-500/month	Included in AWS account	Enable in same region as S3 bucket (ap-south-1)
AWS Textract	ID document OCR for Tier-2 verification	\$0.015/page	\$50-200/month	Included in AWS account	ASYNCR API for multi-page documents
Amazon SES	Transactional email delivery	\$0.10/1000 emails	\$5-50/month	Included in AWS — request production access	Exit SES sandbox: submit support ticket before launch
AWS MediaConvert	Video profile transcoding to HLS	\$0.015/minute of output	\$20-200/month	Included in AWS account	Create MediaConvert IAM role + S3 output bucket
Sentry	Error monitoring + performance tracking	Team: \$26/month	\$26/month	sentry.io — Create org	Add Sentry DSN to all microservices + React Native

Supplementary (P2) — Enhanced Features

Vendor	Purpose	Pricing Model	Est. Monthly Cost	Where to Sign Up	Critical Action
AuthBridge / IDfy	Tier-3 background check API (criminal record)	Per-check: ■50-200	■5K-50K/month	authbridge.com or idfy.com — Enterprise account	Sign MSA + DPA. Pilot with 100 checks before full integration.
Setu (by Pine Labs)	Account Aggregator for income verification (Tier-3)	Per-consent: ■2-10	■1K-10K/month	setu.co — Register as FIU	Requires RBI FIU registration (2-4 weeks). Start early.
HackerOne	Bug bounty programme management	Platform fee: \$2,499/year	\$2,499/year + bounties	hackerone.com — Create programme	Define scope, bounty table, response SLA before going live
Intercom	In-app customer support chat + helpdesk	Starter: \$74/month	\$74-500/month	intercom.com — Create workspace	Add Intercom React Native SDK; connect to admin email
LaunchDarkly	Feature flags for staged rollouts	Starter: \$10/month/seat	\$50-200/month	launchdarkly.com	Optional — can use custom feature_flags DB table initially
Grafana Cloud	Dashboard hosting (Grafana managed)	Free: 50GB logs, 10K metrics	\$0-200/month	grafana.com — Create stack	Alternatively self-host Grafana on ECS (recommended)
Twilio (backup SMS)	SMS OTP fallback if SNS fails	\$0.0075/SMS	\$50-200/month	twilio.com — Account + verify number	Set as fallback in auth-service if SNS error rate > 5%
Cloudflare (optional)	Additional DDoS protection + DNS	Free → \$20/month	\$0-20/month	cloudflare.com — Add site	Can sit in front of CloudFront for extra bot protection

Vendor Contract Checklist

Data Processing Agreement (DPA)	Required for ALL vendors handling personal data. DPDP Act 2023 + GDPR template.
Sub-processor disclosure	Maintain public sub-processor list (required for GDPR). Update within 30 days of change.
Data residency	Confirm where data is stored. Preference: ap-south-1 (Mumbai) for Indian user data.
Incident notification	Vendor must notify within 72 hours of security breach affecting our data.
Right to audit	Include clause allowing us to audit vendor security practices annually.
Data deletion	Vendor must delete all our data within 30 days of contract termination.
SLA & uptime	Minimum 99.9% uptime SLA for P0 vendors. Compensation clauses for breaches.
IP ownership	Confirm all outputs (AI-generated content, processed data) are our property.

04 LEGAL DOCUMENTS FRAMEWORK

Important: This section provides a structural framework and key clauses to brief your legal counsel. Actual legal documents MUST be drafted, reviewed, and approved by a qualified lawyer admitted to practice in India. Do not use this framework as a substitute for professional legal advice. Estimated legal cost: █1-3 lakhs for initial drafting by a tech-specialized firm.

Document 1: Terms of Service

Governs the relationship between the platform and users. Must be agreed to at registration (checkbox + timestamp logged).

1. Eligibility	18+ years only. Sole purpose: seeking a life partner for marriage. Accounts representing deceased persons prohibited.
2. Account Responsibility	One account per person. No sharing. User responsible for all activity under their account.
3. Profile Accuracy	Users warrant all information is truthful. False claims (marital status, profession, income) = grounds for immediate ban.
4. Prohibited Conduct	Scamming, harassment, impersonation, commercial solicitation, content promoting harm, sharing contact details in public profile fields.
5. Content License	User grants platform non-exclusive, royalty-free, worldwide license to display profile content for matchmaking purposes only.
6. Photo & Video Policy	Must be of user only. No celebrities, no nudity, no inappropriate content. Platform may remove without notice.
7. No Guarantee	Platform does not guarantee marriage outcomes. Matching is facilitative; we are not responsible for relationship outcomes.
8. Background Check Disclaimer	Verification badges confirm identity, not character, values, or background beyond stated checks. Users should conduct their own due diligence.
9. Subscriptions & Refunds	Auto-renewing subscriptions. No refunds for unused portion unless legally mandated. Cancellation stops next billing cycle.
10. Limitation of Liability	Platform liability capped at amount paid by user in preceding 3 months. No liability for acts of other users.
11. Dispute Resolution	Arbitration first (India Arbitration and Conciliation Act 1996). Courts of Hyderabad, Telangana have jurisdiction.
12. Termination	Platform may terminate accounts for ToS violations. Users may delete accounts at any time (30-day data retention then deletion).
13. Governing Law	Laws of India. DPDP Act 2023, IT Act 2000, Consumer Protection Act 2019 apply.
14. Changes to ToS	30-day notice for material changes. Continued use = acceptance.

Document 2: Privacy Policy

DPDP Act 2023 (India) + GDPR (for EU/NRI users) compliant. Must have a dedicated Privacy Policy URL. Link in app footer, registration screen, and cookie banner.

Data Controller	[Company Name], [Registered Address], [Email: privacy@domain.com]
Data We Collect	Profile data (name, DOB, religion, caste, photos, bio) Contact data (phone, email) Usage data (interactions, messages) Device data (IP, device ID, OS) Verification documents (stored encrypted, deleted after verification)
Legal Basis (DPDP)	Consent (explicit, withdrawal supported) Legitimate interest (fraud prevention, security) Legal obligation (government requests)
How We Use Data	Matchmaking and recommendations Displaying profiles to other users Safety and fraud prevention Service improvement Legal compliance

Data Sharing	Shared with: other registered users (profile info only) Verified sub-processors (listed at [URL]) Law enforcement (when legally required) NOT sold to advertisers
Retention Periods	Active profiles: while account exists Deleted accounts: 30 days then hard delete Verification docs: 7 days post-verification then deleted Messages: until user deletes Logs: 90 days
User Rights (DPDP)	Access your data Correct inaccurate data Erase your data Withdraw consent Nominate a person to exercise rights on death File complaint with Data Protection Board
Data Localisation	Primary data stored in AWS ap-south-1 (Mumbai). Backups also in ap-south-1.
International Transfers	Claude API (Anthropic, USA) — SCCs in place. Pinecone (USA) — SCCs in place. Full list at [sub-processor URL]
Children's Data	Platform explicitly prohibited for under-18. No knowing collection of minor data. Report to privacy@domain.com if discovered.
Cookies	First-party only: session, preferences. No advertising/tracking cookies. Cookie settings manageable at [URL].
Contact & Complaints	Data Protection Officer: [Name], [Email]. Complaints: Data Protection Board of India [URL when established].
Updates	Material changes communicated via email 30 days in advance. Policy version dated.

Document 3: Cookie Policy

Web app only (React Native app uses secure storage, not cookies). Simple policy sufficient.

Strictly Necessary	Session cookie (JWT), CSRF protection token. Cannot be disabled.
Preferences	Language preference, filter settings, dark/light mode. 1-year expiry.
Analytics	Aggregated page view analytics (self-hosted Plausible or Matomo — no Google Analytics to avoid data export concerns). 30-day retention.
No Third-Party Ads	Zero advertising cookies. No Facebook Pixel, no Google Ads remarketing tags.
Management	Cookie banner on first visit (web). Accept all / Manage / Reject non-essential. Preferences stored in first-party cookie.

Document 4: Community Guidelines

Separate from ToS — plain language, user-facing. Displayed in onboarding and Settings.

Be Honest	Your profile must be genuine. Real photos, real information, real intentions.
Be Respectful	Treat every member with dignity. Rude, offensive, or discriminatory messages will not be tolerated.
Stay Safe	Never share financial information with someone you haven't met. Report suspicious behaviour immediately.
One Person, One Account	Creating multiple accounts to bypass bans or feature limits is prohibited.
No Commercial Use	Advertising, recruitment, promoting services, or any commercial activity on the platform is banned.
Report & Block	Use the Report and Block features. Our safety team reviews every report within 24 hours.
Privacy of Others	Do not share screenshots of conversations or profiles externally without explicit consent.

Compliance Checklist — Before Launch

DPDP Registration	Register as Data Fiduciary with Data Protection Board of India (once operational)
Grievance Officer	Appoint and publish a Grievance Officer name + contact (required under IT Act 2000)
IT Act 2000 — Sec 79	Platform qualifies as intermediary. Maintain due diligence compliance.

Consumer Protection	Register on National Consumer Helpline. Display grievance redressal mechanism.
Lawyer Review	All 4 documents reviewed by tech law firm before app goes live
Legal Version History	All document versions stored with effective dates. Previous versions archived at public URL.
Consent Logging	Log user ToS+PP acceptance: user_id, version, timestamp, IP. Stored in PostgreSQL.
Minor Prevention	Age verification at registration. If underage detected post-registration, immediate account deletion + data erasure.

05 APP STORE & PLAY STORE — ASO STRATEGY

App Store Optimisation (ASO) is the primary organic growth channel for matrimony apps. Telugu matrimony is a high-intent, low-competition keyword cluster vs generic 'matrimony'. Goal: Top 3 ranking for 'Telugu matrimony' on both stores within 6 months of launch. Submit apps at Week 21 (2 weeks review buffer before Week 23 launch).

App Store Listing — iOS (App Store Connect)

App Name (30 chars max)	TeluguMatrimony — Find Match
Subtitle (30 chars max)	Telugu Bride & Groom Search
Category	Primary: Lifestyle Secondary: Social Networking
Age Rating	17+ (mature/suggestive themes — matrimony context)
Primary Language	English (add Telugu as secondary language)
Privacy Nutrition Label	Required: Contact Info User Content Identifiers Usage Data Diagnostics — all collected
App Privacy URL	Link to Privacy Policy page on website
Support URL	Link to support/help page
In-App Purchases	List all: Premium Monthly, Premium Annual, Platinum Monthly, Platinum Annual, Boost, Spotlight, Super Like Pack, Kundli Report

App Store Listing — Google Play (Play Console)

App Name (50 chars max)	TeluguMatrimony - Find Telugu Bride Groom
Short Description (80 chars)	Find your Telugu soulmate. AI-powered matching for AP & TS communities.
Category	Lifestyle
Content Rating	Adults Only 17+ (Dating / Matrimony)
Data Safety Section	Complete all 14 data type disclosures accurately before submission
Target Audience	18-60, not for children (explicitly mark)
Sensitive Permissions	Camera (photo/video), Microphone (VoIP), Location (Near Me search) — justify each in Play Console

Keywords Strategy

iOS: 100-character keyword field. Google Play: Keywords in title + description (indexable). Research via AppFollow, Sensor Tower, or AppTweak before submission.

Primary Keywords (high volume)	telugu matrimony, telugu marriage, telugu bride groom, andhra matrimony, telangana matrimony
Secondary Keywords (medium)	hyderabad matrimony, AP matrimony, TS matrimony, telugu shaadi, telugu wedding
Long-tail Keywords (low comp.)	telugu reddy matrimony, telugu kamma bride, telugu IT groom, andhra NRI matrimony, vizag matrimony
Competitor Brand Adjacent	Do NOT use competitor brand names (trademark violation risk)
Language Keywords	telugu matrimonial sites, telugu pelli, pelli chupulu, telugu vivah
iOS Keyword Field	telugu,matrimony,bride,groom,andhra,telangana,hyderabad,marriage,shaadi,NRI,match,alliance
Google Play Description	Naturally include keywords in first 167 characters of short description and in long description

Screenshots — Required Specification

Both stores require screenshots BEFORE app goes live. Prepare 8 screenshots per device size. Design in Figma. Key principle: show the product, not just marketing text.

Screenshot 1	Daily Matches screen — 'Your AI-matched profiles for today' — show ProfileCard grid with match scores
Screenshot 2	Advanced Search — Filter accordion open showing Telugu community filters, caste selector, 32 filters
Screenshot 3	Profile Detail — Photo carousel + match score 89% + 'Why this match?' bullets — show transparency
Screenshot 4	Chat screen — Show icebreaker panel + scam warning badge + read receipts + AI reply suggestion
Screenshot 5	Verification badges — Show Tier-1/2/3 badge comparison + 'ID Verified' profile card
Screenshot 6	Analytics Dashboard — Weekly stats: views, likes, mutual matches, response rate trend
Screenshot 7	Video Profile — 30-sec reel play screen with privacy options
Screenshot 8	Premium Plans — Plan comparison table — ₹499 vs ₹999 with feature list
Device Sizes (iOS)	6.7" (iPhone 15 Pro Max) + 6.5" (older Plus) — both required. iPad optional.
Device Sizes (Android)	Phone: 1080×1920 minimum. 10-inch tablet: optional but recommended for Play featured.
Localisation	Create Telugu language screenshot set (same screens, Telugu UI) for regional store featuring
Screenshot Text	Keep text overlay short (<5 words). Use Inter Bold. Brand color backgrounds (#1E3A8A or #0A1628).

Preview Video (App Preview / Feature Graphic)

iOS App Preview	15-30 seconds. Captured on device (Xcode screen recording or actual device). No voiceover required. Show 3-5 key flows: search → profile → match → chat.
Google Play Feature Graphic	1024×500px static image. Shown at top of Play listing. Brand navy background + hero tagline + 3 device mockups.
Google Play Promo Video	Link to YouTube video (optional but strongly recommended). 30-60 seconds. Telugu language voiceover explaining 3 USPs.
Video Script (30-sec)	0-5s: Problem ('Finding the right Telugu match is hard') 5-15s: Solution (show AI match + 89% score) 15-25s: Safety (show ID Verified badge + scam protection) 25-30s: CTA ('Download free today')
Production	Can be created in-house using Figma + ScreenFlow/LottieFiles animations. Estimated cost: ₹15-50K outsourced.

Post-Launch ASO Maintenance

Review Responses	Respond to every 1-3 star review within 24 hours. Professional, empathetic tone.
Rating Prompt	In-app prompt after: match accepted + 3 messages sent. iOS: use SKStoreReviewRequest API (Apple-approved). Android: In-App Review API.
Update Cadence	Release at least bi-weekly. Each update: respond to top user complaints. App stores boost recently-updated apps.
A/B Test Listings	Google Play allows A/B testing of screenshots and icon. Test quarterly.
Keyword Monitoring	Monthly keyword rank check via AppFollow or manual search. Adjust keywords in next update.
Competitor Monitoring	Track TeluguMatrimony, TeluguShaadi rankings monthly. React to their new keywords within 30 days.

06 DISASTER RECOVERY PLAN

Defines Recovery Time Objective (RTO) and Recovery Point Objective (RPO) for all tiers. Matrimony platform data is sensitive personal data — data loss is unacceptable. Architecture prioritises: data durability first, availability second, performance third.

RTO / RPO Targets by Tier

Component	RTO (max downtime)	RPO (max data loss)	Tier	Approach
PostgreSQL (primary data)	15 minutes	1 minute	P0 Critical	Multi-AZ + PITR + Read replica
Redis (cache/sessions)	5 minutes	Acceptable loss	P0 Critical	Multi-AZ ElastiCache; sessions re-created
Elasticsearch (search index)	30 minutes	15 minutes	P0 Critical	OpenSearch Serverless auto-replication
S3 (photos/videos)	0 (99.99999999%)	0 (11-9s durability)	P0 Critical	Cross-region replication to ap-southeast-1
ECS Services (app tier)	5 minutes	N/A (stateless)	P0 Critical	Blue/Green deploy; multi-AZ task distribution
Kafka (MSK)	15 minutes	5 minutes	P1 Important	MSK Serverless multi-AZ; consumer offsets in ZK
Pinecone (vector DB)	1 hour	24 hours (re-embed)	P1 Important	Re-embed from PostgreSQL if Pinecone fails
API Gateway / CloudFront	5 minutes	N/A	P0 Critical	AWS-managed, no action needed
LiveKit (call server)	10 minutes	N/A (calls not stored)	P1 Important	EC2 auto-recovery; new instance in 10 min
Claude API (Anthropic)	Graceful degrade	N/A	P1 Important	Fallback to rule-based responses
Razorpay (payments)	1 hour	N/A (Razorpay stores)	P1 Important	Queue failed webhooks; Razorpay dashboard
Admin Panel	2 hours	N/A	P2 Supplementary	Single ECS task; restart from snapshot

Database Backup Strategy

PostgreSQL — RDS Backup Configuration	
Automated Backups	RDS automated backups: enabled. Retention: 35 days (maximum). Storage: S3 in ap-south-1.
Backup Window	Daily: 02:00-03:00 IST (lowest traffic). Maintenance window: Sunday 03:00-04:00 IST.
Point-in-Time Recovery	PITR enabled. Can restore to any second within retention window. Test monthly.
Manual Snapshots	Before every major deployment (database schema migration). Retained indefinitely.
Cross-Region Backup	Weekly snapshot copied to ap-southeast-1 (Singapore) via automated Lambda.
Backup Encryption	KMS-encrypted. Same KMS key as production data. Key rotation: annual.
Restore Testing	Monthly restore drill: restore latest backup to staging environment. Verify data integrity. Document RTO achieved.
pg_dump (logical backup)	Weekly full pg_dump to S3. Retained 90 days. Faster than RDS snapshot for individual table recovery.
Redis — Backup Configuration	
AOF (Append-Only File)	Enabled: appendfsync=everysec. Trades slight performance for 1-second RPO.
Snapshot (RDB)	Daily RDB snapshot at 03:00 IST. Stored in S3 (ElastiCache managed).
Acceptable Loss Strategy	Most Redis data (session tokens, rate limit counters) is transient. On full failure: sessions expire, users re-login. Acceptable for most keys.

Critical Keys to Protect	Conversation encryption key references (KMS key IDs). These back PostgreSQL messages table so loss is recoverable from DB.
S3 Media — Backup & Durability	
Native Durability	S3 Standard: 99.999999999% (11 nines) durability. Photos are inherently safe.
Versioning	S3 versioning enabled on all media buckets. Accidental deletion recoverable for 90 days.
Cross-Region Replication	CRR: ap-south-1 → ap-southeast-1 (Singapore). RTO for regional outage: 0 (Singapore serves while Mumbai recovers).
Lifecycle Rules	Standard → Standard-IA after 30 days (media not accessed). Standard-IA → Glacier after 1 year. Glacier after 7 years → deletion.
Object Lock	Verification documents: WORM (Write Once Read Many) lock for 7 days post-verification. Prevent accidental deletion during legal hold.

Failover Procedures

Scenario 1: RDS Primary Database Failure

Expected: AWS Multi-AZ automatic failover. RTO: < 2 minutes (DNS propagation). Manual steps only if auto-failover fails.

- 1. CloudWatch alarm fires: RDS events → SNS → PagerDuty → on-call engineer
- 2. Check RDS console: confirm Multi-AZ failover initiated automatically
- 3. If auto-failover not triggered within 3 minutes: AWS Console → RDS → Failover (manual button)
- 4. Verify application reconnects: check ECS task logs for DB connection errors
- 5. If ECS services show DB errors after failover: rolling restart of all ECS services (forces new connections)
- 6. Confirm write operations working: test POST /profiles from admin panel
- 7. Post-incident: RCA within 24 hours. Review Multi-AZ health. Add monitoring for failover latency.

Scenario 2: Full AWS ap-south-1 (Mumbai) Region Outage

Rare but possible. Cross-region warm standby in ap-southeast-1 (Singapore). Target RTO: 2 hours.

- 1. PagerDuty: all health checks fail simultaneously → escalate to Super Admin immediately
- 2. AWS Status page confirms regional outage: activate cross-region DR plan
- 3. Singapore region: promote RDS read replica to primary (AWS Console → Promote)
- 4. Update Route 53: change API endpoint DNS from Mumbai ALB to Singapore ALB (TTL: 60 seconds)
- 5. Singapore ECS services: scale up from 0 to production task counts (Terraform apply -var='region=ap-southeast-1')
- 6. CloudFront: already serving S3 from Singapore replica (CRR) — no change needed
- 7. Kafka: MSK in Singapore had consumers paused — restart consumers against Singapore MSK cluster
- 8. Anthropic API: no change (global endpoint). Razorpay: no change (their infra).
- 9. Notify users via email (SES in Singapore): 'Brief maintenance in progress. We'll be back shortly.'
- 10. Mumbai restoration: when Mumbai recovers, reverse DNS, sync data delta, decommission Singapore primary.

Scenario 3: Claude API Outage (Anthropic Down)

- 1. ai-service detects Anthropic API error rate > 10% → switches to FALLBACK MODE (circuit breaker)
- 2. Bio generator: show 3 profession-specific template bios from database instead of generated ones
- 3. Match explanation: show top-3 structured factors from XGBoost scores (no narrative text)
- 4. Scam detection: fall back to rule-based keyword matching (pre-defined scam phrase list in Redis)
- 5. Icebreakers: serve pre-written icebreaker templates matched by shared interests (no Claude)
- 6. Auto-translate: disable button, show 'Translation temporarily unavailable'
- 7. Smart search: disable AI search mode, show notice to user
- 8. Circuit breaker resets: retry Claude every 5 minutes. Auto-restore when error rate < 1%.

Scenario 4: Data Breach / Security Incident

DPDP Act 2023 requires notification to Data Protection Board within 72 hours of breach discovery.

- 1. Discovery: GuardDuty alert OR internal detection OR external report → immediately alert CISO/Super Admin
- 2. First 30 minutes: isolate affected system (revoke IAM keys, remove from load balancer, capture logs)
- 3. Scope assessment: what data, how many users, exfiltration confirmed or suspected?
- 4. Hour 1-4: Engage AWS Security Response (Business Support) + legal counsel
- 5. Hour 4-24: Forensic investigation. Preserve all logs (set S3 Object Lock on relevant log files)
- 6. Hour 24-48: Draft breach notification for Data Protection Board (when operational) + affected users
- 7. Hour 48-72: Submit formal notification. Legal counsel reviews before submission.
- 8. User notification: email all affected users within 72 hours with: what happened, what data, what we're doing
- 9. Post-breach: full security audit, penetration test, remediation within 30 days, public transparency report

Monitoring & Alerting Stack

CloudWatch Alarms	RDS CPU > 80% Redis memory > 85% ECS task count < desired API error rate > 5% P99 latency > 3s
PagerDuty Escalation	Severity 1 (platform down): call engineer immediately (24×7). Severity 2 (degraded): page within 5 min. Severity 3 (minor): email + Slack.
On-Call Rotation	Primary + secondary on-call. Rotate weekly. Primary must respond within 5 min of S1 page.
Grafana Dashboards	Platform health (real-time) Business metrics (daily) Claude API cost (hourly) Security events (real-time)
Synthetic Monitoring	Every 5 minutes: automated E2E test (register → search → like). Failure → S2 alert.
Log Aggregation	CloudWatch Logs Insights: all service logs. Retention: 90 days. Critical errors → SNS → Slack.
Uptime Status Page	Public status page (Statuspage.io or self-hosted) at status.domain.com. Update within 15 min of confirmed incident.
Chaos Testing	Monthly: randomly terminate 1 ECS task (simulated failure). Verify auto-recovery. No user impact expected.

DR Test Schedule

Test	Frequency	Duration	Conducted By	Success Criteria
RDS PITR restore to staging	Monthly	30 min	DevOps	Data restored correctly to specified timestamp
ECS task failure & auto-recovery	Monthly	10 min	DevOps (automated chaos)	Service restores within 5 min, no user errors
Redis flush & application recovery	Quarterly	20 min	DevOps	App serves users; sessions re-created gracefully
Cross-region DNS failover drill	Quarterly	2 hours	Full team	Singapore serves traffic; RTO < 2 hours achieved
Claude API circuit breaker test	Monthly	15 min	Backend team	Fallback responses served correctly for all 9 features
Full DR simulation	Annually	Full day	All teams	End-to-end recovery from simulated regional outage
Breach response tabletop	Semi-annually	3 hours	Legal + Engineering + Management	Team knows roles; notification drafted within 48hr

SUPPLEMENTARY SPECIFICATION v3.1 — COMPLETE

All 6 pending modules are now fully documented. Together with the v3.0 Main Spec, this constitutes the complete production-ready blueprint for the Matrimony Platform. Share both documents with engineering, legal, and operations teams before kickoff. No further known gaps remain. Ready to build.